NAYANA PULSE

HACKFEST - 2026

Team Name:	[Your Team Name]
College Name:	[Your College Name], [State]
Track:	Healthcare
Problem Statement:	Millions of paralyzed patients (ALS, stroke) cannot communicate or monitor health easily. Nayana Pulse uses a webcam for eye-tracking communication and non-contact health monitoring.

Approach:

Pure-software solution using standard laptop webcams.

Eye-tracking for cursor control and quadrant-based communication.

Computer vision for non-contact vitals (blink rate, head tilt).

Technology Stack:

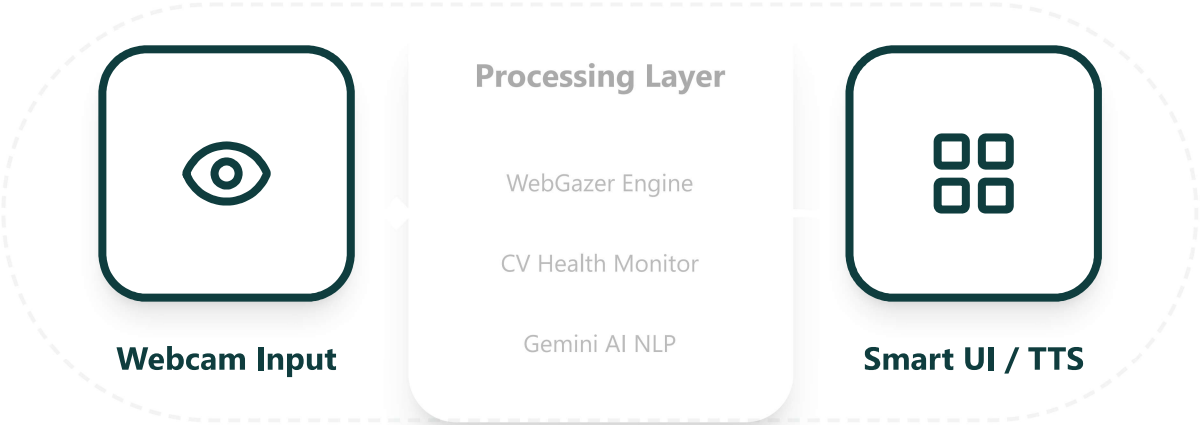
FRONTEND
React.js & Tailwind

EYE TRACKING
WebGazer.js

AI ENGINE

SPEECH

Architecture:



Additional Information:

Scalability & Impact

Nayana Pulse is designed for global accessibility. With $0 hardware requirement beyond a basic laptop, it can be deployed in rural clinics and low-resource hospitals instantly.

Winning Pitch Strategy

1. **The Silence:** Start the pitch without speaking to build curiosity